MINI PROJECT

Secure Campus Network Using Cisco Packet Tracer

Topics: Router Configuration • Switch Security • SSH • Port Security • Console/Enable Security • Password Encryption

1. Project Objective

Design and implement a small secure campus network in Cisco Packet Tracer. The project introduces students to basic Cisco device configuration and security mechanisms, including router and switch configuration, console and privileged EXEC security, SSH remote access, switch port security, and password encryption.

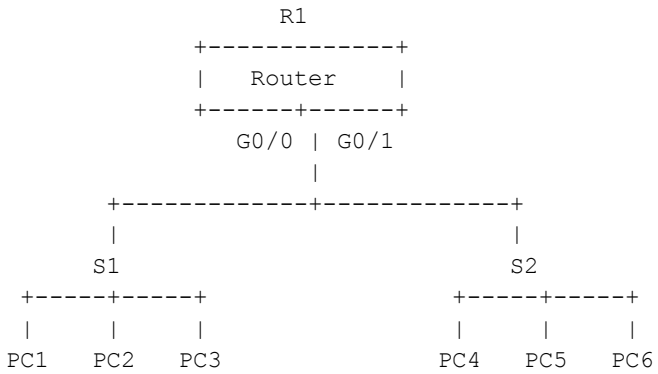
2. Network Scenario

A small university department has one router, two Cisco switches, and six PCs. The router connects two IP networks. The switches provide LAN connectivity and are secured against unauthorized access and unauthorized devices.

3. Devices Required

- 1 × Cisco 2911 Router (R1)
- 2 × Cisco 2960 Switches (S1 and S2)
- 6 × PCs (PC1–PC6)
- Copper straight-through cables

4. Network Topology




```
Network 1: 192.168.10.0/24
Network 2: 192.168.20.0/24
```

5. IP Addressing Plan

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.10.1	255.255.255.0	—
R1	G0/1	192.168.20.1	255.255.255.0	—

S1	VLAN 1	192.168.10.2	255.255.255.0	192.168.10.1
S2	VLAN 1	192.168.20.2	255.255.255.0	192.168.20.1
PC1	NIC	192.168.10.10	255.255.255.0	192.168.10.1
PC2	NIC	192.168.10.11	255.255.255.0	192.168.10.1
PC3	NIC	192.168.10.12	255.255.255.0	192.168.10.1
PC4	NIC	192.168.20.10	255.255.255.0	192.168.20.1
PC5	NIC	192.168.20.11	255.255.255.0	192.168.20.1
PC6	NIC	192.168.20.12	255.255.255.0	192.168.20.1

6. Physical Connections

- R1 G0/0 → S1 G0/1
- R1 G0/1 → S2 G0/1
- S1 Fa0/1 → PC1
- S1 Fa0/2 → PC2
- S1 Fa0/3 → PC3
- S2 Fa0/1 → PC4
- S2 Fa0/2 → PC5
- S2 Fa0/3 → PC6

7. Task 1 – Configure Router R1

Router Configuration

```
enable
configure terminal
hostname R1

interface gigabitEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
exit

show ip interface brief
```

8. Task 2 – Secure Router Console and Enable Mode

Console and Privileged EXEC Security

```
line console 0
password Console@123
login
exit

enable secret Enable@123
```

The console password protects console access. The enable secret protects privileged EXEC mode (R1#). Using enable secret is preferred over a plain enable password.

9. Task 3 – Password Encryption

Enable Password Encryption

```
service password-encryption
show running-config
```

service password-encryption causes line passwords configured in the device configuration to be stored in an obfuscated form. It is not equivalent to strong cryptographic protection; enable secret should be used for privileged access.

10. Task 4 – Configure SSH on R1

SSH Configuration

```
ip domain-name campus.local
username admin privilege 15 secret Admin@123
crypto key generate rsa
! Enter 1024 when prompted (or 2048 if supported)
ip ssh version 2

line vty 0 4
login local
transport input ssh
exit

show ip ssh
```

11. Task 5 – Configure Switch S1

Basic S1 Configuration

```
enable
configure terminal
hostname S1
enable secret Enable@123
service password-encryption

interface vlan 1
ip address 192.168.10.2 255.255.255.0
no shutdown
exit

ip default-gateway 192.168.10.1
```

12. Task 6 – Secure S1 Console and Configure SSH

S1 Console and SSH

```
line console 0
password SwitchConsole@123
login
exit

ip domain-name campus.local
username admin privilege 15 secret Admin@123
```

```
crypto key generate rsa
! Enter 1024 when prompted (or 2048 if supported)
ip ssh version 2

line vty 0 4
login local
transport input ssh
exit
```

13. Task 7 – Configure Port Security on S1

Secure PC Ports

```
interface fastEthernet 0/1
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
exit

interface fastEthernet 0/2
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
exit

interface fastEthernet 0/3
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
exit
```

14. Task 8 – Configure Switch S2

Basic S2 Configuration

```
enable
configure terminal
hostname S2
enable secret Enable@123
service password-encryption

interface vlan 1
ip address 192.168.20.2 255.255.255.0
no shutdown
exit

ip default-gateway 192.168.20.1

line console 0
password SwitchConsole@123
login
```

```
exit

ip domain-name campus.local
username admin privilege 15 secret Admin@123
crypto key generate rsa
! Enter 1024 when prompted (or 2048 if supported)
ip ssh version 2

line vty 0 4
login local
transport input ssh
exit
```

15. Task 9 – Configure Port Security on S2

Secure PC Ports

```
interface range fastEthernet 0/1-3
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
exit
```

16. Task 10 – Configure the PCs

Configure each PC using Desktop → IP Configuration.

PC	IP Address	Subnet Mask	Gateway
PC1	192.168.10.10	255.255.255.0	192.168.10.1
PC2	192.168.10.11	255.255.255.0	192.168.10.1
PC3	192.168.10.12	255.255.255.0	192.168.10.1
PC4	192.168.20.10	255.255.255.0	192.168.20.1
PC5	192.168.20.11	255.255.255.0	192.168.20.1
PC6	192.168.20.12	255.255.255.0	192.168.20.1

17. Task 11 – Test Network Connectivity

From PC1 Command Prompt

```
ping 192.168.10.1
ping 192.168.20.10
ping 192.168.20.11
ping 192.168.20.12
```

Expected result: successful replies after ARP/MAC learning has completed.

18. Task 12 – Test SSH Access

From PC1

```
ssh -l admin 192.168.10.2
Password: Admin@123
```

```
ssh -l admin 192.168.10.1
Password: Admin@123
```

From PC4

```
ssh -l admin 192.168.20.2
Password: Admin@123
```

19. Task 13 – Verify Port Security

Verification Commands

```
show port-security
show port-security address
show port-security interface fastEthernet 0/1
show mac address-table
```

Students should verify that port security is enabled, the maximum secure MAC address count is 1, sticky MAC learning is active, and the violation mode is shutdown.

20. Task 14 – Demonstrate a Port Security Violation

First connect PC1 to S1 Fa0/1 and generate traffic, for example by pinging the router. The switch learns PC1's MAC address. Disconnect PC1 and connect another PC to Fa0/1. Because only one secure MAC address is permitted, the switch should detect a violation.

Check the Port

```
show port-security interface fastEthernet 0/1
```

Recover the Interface

```
configure terminal
interface fastEthernet 0/1
shutdown
no shutdown
exit
```

21. Task 15 – Disable Unused Switch Ports

S1 and S2

```
interface range fastEthernet 0/4-24
shutdown
exit
```

This reduces the risk of unauthorized devices being connected to unused active ports.

22. Task 16 – Save the Configuration

Run on R1, S1 and S2

```
copy running-config startup-config
```

23. Final Verification Checklist

Test	Expected Result	Student Result
PC1 → Router ping	Successful	
PC1 → PC4 ping	Successful	
SSH → R1	Successful	
SSH → S1	Successful	
SSH → S2	Successful	
Port security	Enabled	
Sticky MAC learning	Successful	
Unauthorized MAC test	Violation detected	
Unused ports	Shutdown	

24. Student Deliverables

- Packet Tracer (.pkt) file
- Network topology screenshot
- IP addressing table
- Router and switch configuration screenshots
- SSH login screenshot
- Port-security verification screenshot
- Port-security violation demonstration screenshot
- Connectivity test screenshots
- Short project report with observations and conclusion

25. Project Evaluation – 100 Marks

Component	Marks
Network topology	10
Router configuration	10
Switch configuration	10
Console security	10
Enable-mode security	10
Password encryption	10
SSH configuration	15
Port security	15
Testing & troubleshooting	5
Documentation/presentation	5
Total	100

27. Conclusion

This mini project provides practical experience in securing Cisco routers and switches. Students configure basic IP connectivity, protect console and privileged EXEC access, enable SSH for secure remote administration, encrypt locally configured passwords, and use switch port security to restrict unauthorized devices. The completed Packet Tracer topology demonstrates how multiple basic security controls can be combined to improve the security of a small campus network.